# Hypergamy Is Solved

Feasibility, Deficit, and Equilibrium in Status-Ordered Matching Markets

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**Abstract**

Hypergamy, the tendency to form unions with a partner of higher status, is usually argued about as a behavioral claim. We instead model the marriage market as a coupling problem between Borel probability measures on Polish type spaces and show that, absent taste heterogeneity, every aggregate question about hypergamy reduces to a question about two marginal distributions. We prove: (i) universal hypergamy is feasible if and only if one status distribution first-order stochastically dominates the other (Strassen); (ii) otherwise the minimal unmatched mass is the *Hypergamy Deficit* $\mathsf{D} = \sup_t (G_{\mathfrak{W}} - G_{\mathfrak{M}})^+$ (a continuum Hall marriage theorem); (iii) no behavioral strategy changes any of this (*Cope Invariance*), and status gains are zero-sum in rank (*Conservation of Rank*); (iv) under status-ranked preferences the unique Gale–Shapley stable matching is positive assortative, and under Becker–Shapley–Shubik transferable utility it is the comonotone optimal transport plan, with the Choo–Siow entropic relaxation tending to random matching as taste shocks grow. For equal height variances we derive a closed-form deficit for required height gaps, $\mathsf{D}(\delta) = (2\Phi(\frac{\delta - \Delta m}{2\sigma}) - 1)^+$, which vanishes up to $\Delta m \approx 14$ cm and, at the pooled US standard deviation $\sigma \approx 7.35$ cm, reaches 0.74 at one foot. With several traits, checklist requirements shrink the eligible pool multiplicatively when traits are independent, trade-offs reduce to a scalar index, and a required rank gap $\delta$ leaves exactly $\delta$ of one side unmatched. We model partner history as a comparison effect, which applies to both sexes, and show that one ex remembered a standard deviation better than they were costs more than four ordinary ones.

## 1 The Marriage Market as a Measure Space

**Definition 1.1** (Market)**.** A *marriage market* is a tuple $(\mathcal{X}, \mathcal{Y}, \mu, \nu, s)$ where $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces of agent types for the two sides $\mathfrak{M}$ and $\mathfrak{W}$, $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$ are Borel probability measures describing the populations, and $s : \mathcal{X} \sqcup \mathcal{Y} \to \mathbb{R}$ is a measurable *status functional* (height, income, education, follower count, or any scalar index thereof). We write $\mu_s = s_{\#}\mu$ and $\nu_s = s_{\#}\nu$ for the pushforward status distributions, with CDFs $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ and survival functions $G_{\mathfrak{M}}(t) = \mu(s \geq t)$, $G_{\mathfrak{W}}(t) = \nu(s \geq t)$.

**Definition 1.2** (Matching)**.** A *matching* is a coupling $\pi \in \Pi(\mu, \nu)$, i.e. a Borel probability measure on $\mathcal{X} \times \mathcal{Y}$ with marginals $\mu$ and $\nu$. A *partial matching* is a sub-coupling: a measure $\pi$ with $(p_1)_{\#}\pi \leq \mu$ and $(p_2)_{\#}\pi \leq \nu$. Its *mass* $|\pi| = \pi(\mathcal{X} \times \mathcal{Y})$ is the amount of each side that is matched.

**Definition 1.3** (Hypergamy)**.** A partial matching $\pi$ is *hypergamous for* $\mathfrak{W}$ if $\pi(\{s(x) \geq s(y)\}) = |\pi|$, and *strictly hypergamous* if the inequality is strict $\pi$-a.e. The *hypergamy index* is $\mathsf{H}(\pi) = \pi(\{s(x) > s(y)\})$ and the *hypogamy index* is $\mathsf{H}^{\downarrow}(\pi) = \pi(\{s(x) < s(y)\})$.

*Remark* 1.4. Every definition is symmetric under exchanging $(\mathcal{X}, \mu)$ with $(\mathcal{Y}, \nu)$. Conventional usage applies hypergamy to $\mathfrak{W}$, and we follow that convention for notation only. No result depends on which side is labeled which, and the reader is free to swap the letters if it makes them feel better.

## 2 Who Can Marry Up? Feasibility and the Deficit

**Theorem 2.1** (Hypergamy Feasibility)**.** *A hypergamous matching $\pi \in \Pi(\mu, \nu)$ exists if and only if $\mu_s$ first-order stochastically dominates $\nu_s$:*

$$G_{\mathfrak{M}}(t) \geq G_{\mathfrak{W}}(t) \qquad \text{for all } t \in \mathbb{R}. \tag{1}$$

*Proof.* ($\Rightarrow$) If $\pi$ is hypergamous then $\{s(y) \geq t\} \subseteq \{s(x) \geq t\}$ up to a $\pi$-null set, so $G_{\mathfrak{W}}(t) = \pi(s(y) \geq t) \leq \pi(s(x) \geq t) = G_{\mathfrak{M}}(t)$. ($\Leftarrow$) With generalized inverses $F^{-1}(u) = \inf\{t : F(t) \geq u\}$, condition (1) gives $F^{-1}_{\mathfrak{M}}(u) \geq F^{-1}_{\mathfrak{W}}(u)$ for all $u$, so for $U \sim \mathrm{Unif}(0,1)$ the pair $(F^{-1}_{\mathfrak{M}}(U), F^{-1}_{\mathfrak{W}}(U))$ couples $(\mu_s, \nu_s)$ on $\{a \geq b\}$. Lift to $\mathcal{X} \times \mathcal{Y}$ by disintegrating $\mu, \nu$ along the fibers of $s$ (possible since the spaces are Polish). This is the one-dimensional case of Strassen's theorem [18]. □

When (1) fails, someone has to marry down or stay single. We compute exactly how many. Allow general masses $\mu(\mathcal{X}) = m$, $\nu(\mathcal{Y}) = w$ and let $\mathcal{C} = \{(x, y) : s(x) \geq s(y)\}$.

**Theorem 2.2** (Hypergamy Deficit Formula)**.** *The maximal mass of a partial matching supported on $\mathcal{C}$ is*

$$\max_{\pi(\mathcal{C}^c)=0} |\pi| = w - \mathsf{D}, \qquad \mathsf{D} := \sup_{t \in \mathbb{R}} \big(\nu(s \geq t) - \mu(s \geq t)\big)^+. \tag{2}$$

$\mathsf{D}$ *is the minimal mass of $\mathfrak{W}$ left unmatched if every union is hypergamous.*

*Proof. Upper bound.* Agents in $\mathfrak{W}$ with $s(y) \geq t$ are compatible only with $\{x : s(x) \geq t\}$, so at least $\nu(s \geq t) - \mu(s \geq t)$ of them are unmatched; take the supremum. *Achievability.* Leave the top $\mathsf{D}$ of $\mathfrak{W}$ by status unmatched, splitting an atom if necessary, and use only the top $w - \mathsf{D} \leq m$ of $\mathfrak{M}$. These women and men have survival functions $(G_{\mathfrak{W}} - \mathsf{D})^+$ and $\min(G_{\mathfrak{M}}, w - \mathsf{D})$, and the second dominates the first because $G_{\mathfrak{W}} - G_{\mathfrak{M}} \leq \mathsf{D}$ and $G_{\mathfrak{W}} \leq w$, so Theorem 2.1, rescaled to mass $w - \mathsf{D}$, matches them hypergamously. For finitely supported measures this is the deficiency form of Hall's theorem: compatibility is *nested*, so Hall's condition need only be checked on upper sets. □

**Corollary 2.3.** $\mathsf{D} = 0$ *iff* (1) *holds (when $m = w$), and $\mathsf{D} \geq (w - m)^+$. A skewed sex ratio is a lower bound on the deficit; everything above it is caused by the shape of the distributions.*

**Example 2.4.** Let $\mu_s = \mathrm{Unif}[0,1]$ and $\nu_s = \mathrm{Beta}(2,1)$, so $G_{\mathfrak{M}}(t) = 1 - t$ and $G_{\mathfrak{W}}(t) = 1 - t^2$. Then $\mathsf{D} = \sup_t (t - t^2) = \frac{1}{4}$ at $t^* = \frac{1}{2}$. With a perfectly balanced sex ratio, one quarter of $\mathfrak{W}$ still cannot be matched hypergamously, and the bottleneck sits exactly at the median of $\mathfrak{M}$.

## 3 Application: The Six-Foot Rule

Height is the cleanest test case: scalar, observable, approximately Gaussian within each sex, and the subject of strongly held opinions. Take $s =$ height in cm with $\mu_s = N(m_{\mathfrak{M}}, \sigma^2_{\mathfrak{M}})$ and $\nu_s = N(m_{\mathfrak{W}}, \sigma^2_{\mathfrak{W}})$. We calibrate to US adult means from NHANES 2015–2018 [12], $m_{\mathfrak{M}} \approx 175.3$ and $m_{\mathfrak{W}} \approx 161.3$, write $\Delta m \approx 14.0$, and take $\sigma_{\mathfrak{M}} = 7.6$ and $\sigma_{\mathfrak{W}} = 7.1$. The report gives percentiles rather than standard deviations, and its percentiles imply 7.4–7.5 and 6.9–7.1.

**Proposition 3.1** (The $-37.5$ cm Theorem)**.** *For Gaussian marginals with $\sigma_{\mathfrak{M}} \neq \sigma_{\mathfrak{W}}$, the survival functions cross exactly once, at*

$$t_\times = \frac{\sigma_{\mathfrak{M}} m_{\mathfrak{W}} - \sigma_{\mathfrak{W}} m_{\mathfrak{M}}}{\sigma_{\mathfrak{M}} - \sigma_{\mathfrak{W}}} \approx -37.5\,\mathrm{cm}. \tag{3}$$

*Hence* (1) *fails only on $(-\infty, t_\times)$ and $\mathsf{D}(0) < 10^{-150}$.*

*Proof.* $G_{\mathfrak{M}}(t) = G_{\mathfrak{W}}(t)$ iff $(t - m_{\mathfrak{M}})/\sigma_{\mathfrak{M}} = (t - m_{\mathfrak{W}})/\sigma_{\mathfrak{W}}$, a linear equation with unique root $t_\times$. Above it, $\mathfrak{M}$ has the smaller standardized score, so $G_{\mathfrak{M}} > G_{\mathfrak{W}}$. Below it the gap is at most $\Phi$ of a standardized value near $-28$. □

*Remark* 3.2. "He must be taller than me" is therefore feasible for every human of positive height. The only obstruction occurs below $-37.5$ cm, a region the author was unable to sample. Other readings of the report's percentiles move the crossing anywhere from $-6$ to $-108$ cm, which the author did not sample either.

Now require a gap. A union satisfies the *$\delta$-rule* if $s(x) \geq s(y) + \delta$. Shifting $\nu_s$ by $\delta$ reduces this to Theorem 2.2, so $\mathsf{D}(\delta) = \sup_t (G_{\mathfrak{W}}(t) - G_{\mathfrak{M}}(t + \delta))^+$.

**Theorem 3.3** (Closed-Form Height Deficit)**.** *If $\sigma_{\mathfrak{M}} = \sigma_{\mathfrak{W}} = \sigma$, then*

$$\mathsf{D}(\delta) = \left( 2\,\Phi\!\left( \frac{\delta - \Delta m}{2\sigma} \right) - 1 \right)^+ . \tag{4}$$

$\mathsf{D}$ *vanishes iff $\delta \leq \Delta m$, and is concave increasing beyond it with initial slope $(\sigma\sqrt{2\pi})^{-1}$, or* 5.4% *per* cm *at the pooled $\sigma = 7.35$* cm*.*

*Proof.* With $z = (t - m_{\mathfrak{W}})/\sigma$ and $c = (\delta - \Delta m)/\sigma$, the gap is $\Phi(z + c) - \Phi(z)$, the Gaussian mass of an interval of length $c$. For $c > 0$ it is maximized by centering the interval at the mode, $z = -c/2$, giving $2\Phi(c/2) - 1$. Its $\delta$-derivative $\varphi(c/2)/\sigma$ is decreasing in $c$. □

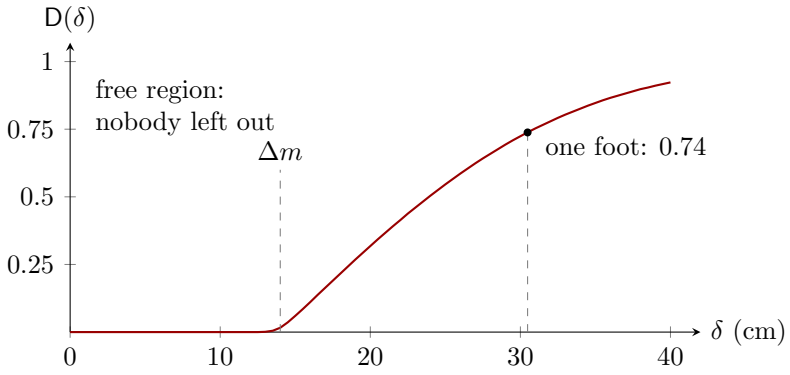



Figure 1: Minimal fraction of $\mathfrak{W}$ left unmatched as a function of the required height gap, US calibration with unequal variances. Essentially zero until about 13 cm, then roughly 5% per centimeter.

**Theorem 3.4** (The Six-Foot Theorem)**.** *Suppose a fraction $\alpha$ of $\mathfrak{W}$ requires $s(x) \geq \tau$, and everyone else imposes any constraint whatsoever. Then at least $(\alpha - G_{\mathfrak{M}}(\tau))^+$ of $\mathfrak{W}$ is unmatched in every feasible matching. For $\tau = 182.9$* cm *(six feet), $G_{\mathfrak{M}}(\tau) \approx 0.16$.*

*Proof.* The constrained set $A$ has neighborhood inside $\{s \geq \tau\}$, of mass $G_{\mathfrak{M}}(\tau)$. Apply the Hall bound from Theorem 2.2. □

*Remark* 3.5. Theorem 3.4 holds regardless of personality, effort, or app. It is also generous: it assumes every six-footer is available and interested.

# 4 The Blackpill Theorems

**Lemma 4.1** (Conservation of Status)**.** *If $s \in L^1(\mu) \cap L^1(\nu)$, then for every $\pi \in \Pi(\mu,\nu)$, $\int (s(x) - s(y))\, d\pi = \int t\, d\mu_s - \int t\, d\nu_s$. The average status gap across all couples does not depend on who marries whom.*

*Proof.* Integrate against $\pi$ and use its marginals. □

**Definition 4.2** (Strategy)**.** A *strategy* is any map $\mathfrak{S} : \Pi(\mu,\nu) \to \Pi(\mu,\nu)$: every intervention that changes who matches with whom without changing the population. This includes openers, dating apps, "just be confident," and every other object the literature calls *cope*.

**Theorem 4.3** (Cope Invariance)**.** *If $\Psi : \Pi(\mu,\nu) \to \mathbb{R}$ depends on $\pi$ only through its marginals, then $\Psi \circ \mathfrak{S} = \Psi$ for every strategy $\mathfrak{S}$. In particular the mean status gap, the feasibility of universal hypergamy, the deficit $\mathsf{D}$, and the equilibrium index $\mathsf{H}^*$ of (8) are invariant under every strategy.*

*Proof.* Strategies preserve marginals, and each listed quantity is a functional of $(\mu_s, \nu_s)$ alone. □

**Corollary 4.4** (It's Over)**.** *If $\mu_s = \nu_s$, no strictly hypergamous matching exists and no strategy produces one: by Lemma 4.1 applied to the bounded status* $\arctan s$*, the integrand* $\arctan s(x) - \arctan s(y)$ *would be positive $\pi$-a.e. with zero integral.*

The only escape is to change the marginals, replacing $s$ by $s + h$ with $h \geq 0$ on a subset of agents. We call this *maxxing*.

**Theorem 4.5** (Conservation of Rank)**.** *Let $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ be continuous and let the market clear at the comonotone coupling. If $U = F_{\mathfrak{M}}(s(x))$ is an agent's own quantile and $V = F_{\mathfrak{W}}(s(y))$ their partner's, then $V = U$ a.s. and $U \sim \mathrm{Unif}(0,1)$. Hence for any perturbation $s \mapsto s + h$ that keeps $F_{\mathfrak{M}}$ and $F_{\mathfrak{W}}$ continuous, maxxing included,*

$$\mathbb{E}[V] = \tfrac{1}{2}, \qquad \mathrm{Var}(V) = \tfrac{1}{12} \tag{5}$$

*before and after. Maxxing by a subpopulation raises its partners' quantiles by exactly the amount it lowers everyone else's.*

*Proof.* Under the comonotone coupling $s(y) = F_{\mathfrak{W}}^{-1}(F_{\mathfrak{M}}(s(x)))$, so $V = U$. The probability integral transform gives $U \sim \mathrm{Unif}(0,1)$ for any continuous $F_{\mathfrak{M}}$, perturbed or not. □

> **Remark (The Blackpill, Stated Precisely).** Theorems 4.3 and 4.5 are the rigorous content of "it's over": aggregate outcomes are pinned by the marginals, strategies act within a fiber of fixed marginals, and status gains are transfers of rank, not creation of it. But both are statements about the *population*. For an individual at quantile $u$ the comonotone map is strictly increasing, so raising your own status raises your partner's quantile one-for-one, and for $\theta > 0$ (Section 7) outcomes are random rather than determined.
>
> **It is over for the population. It is not over for you.**

# 5 Everyone Wants Up, Almost Everyone Goes Down

Take finite sides $|\mathfrak{M}| = |\mathfrak{W}| = n$ with distinct statuses and *status-ranked* preferences: every agent strictly prefers a higher-status partner, and any partner to none. A matching is *stable* if no pair would both rather be with each other than with their assigned partners.

**Theorem 5.1** (Uniqueness of the Assortative Equilibrium)**.** *Under status-ranked preferences the unique stable matching pairs the $k$-th highest of $\mathfrak{M}$ with the $k$-th highest of $\mathfrak{W}$ for every $k$, and Gale–Shapley deferred acceptance [13] returns it whichever side proposes.*

*Proof.* Induct on $n$. If the two top agents are not matched to each other, they form a blocking pair. Remove them; the reduced market has the same structure. □

**Corollary 5.2** (Everyone Wants Up, Almost Everyone Goes Down)**.** *In Example 2.4 the assortative matching is the quantile map $T = F_{\mathfrak{M}}^{-1} \circ F_{\mathfrak{W}}$, $T(b) = b^2 < b$ on $(0,1)$, and no pair blocks it: a man at quantile $u$ prefers a woman at quantile $v$ only if $v > u$, and she prefers him only if $u > v$. Hence $\mathsf{H} = 0$ and $\mathsf{H}^{\downarrow} = 1$: every member of $\mathfrak{W}$ prefers to marry up, the market is stable, and almost every member of $\mathfrak{W}$ marries down, by an average of $\int_0^1 (b - b^2)\, 2b\, db = \frac{1}{6}$.*

## 6 Transferable Utility and Optimal Transport

Following Becker [4] and Shapley–Shubik [15], a couple with statuses $a$ and $b$ generates surplus $f(a,b)$, which partners may split. The equilibrium solves the Monge–Kantorovich problem

$$\sup_{\pi \in \Pi(\mu,\nu)} \int f(s(x), s(y))\, d\pi = \inf_{u(x)+v(y) \geq f(s(x),s(y))} \left\{ \int u\, d\mu + \int v\, d\nu \right\}, \tag{6}$$

where equality is Kantorovich duality and $(u, v)$ are the equilibrium utilities [20].

**Theorem 6.1** (Becker via Cyclical Monotonicity)**.** *If $f$ is strictly supermodular ($\partial_{ab} f > 0$), every optimal plan has status projection equal to the comonotone coupling $\gamma^* = (F_{\mathfrak{M}}^{-1}, F_{\mathfrak{W}}^{-1})_{\#} \operatorname{Unif}(0,1)$.*

*Proof.* Optimal plans are cyclically monotone. If $a_1 < a_2$ and $b_1 > b_2$ were both in the support, then

$$f(a_2, b_1) + f(a_1, b_2) - f(a_1, b_1) - f(a_2, b_2) = \int_{a_1}^{a_2} \int_{b_2}^{b_1} \partial_{ab} f\, d\beta\, d\alpha > 0, \tag{7}$$

so swapping partners strictly improves total surplus, a contradiction. □

So bargaining or not, the realized hypergamy index is the same:

$$\mathsf{H}^* := \mathsf{H}(\gamma^*) = \operatorname{Leb}\{u \in (0,1) : F_{\mathfrak{M}}^{-1}(u) > F_{\mathfrak{W}}^{-1}(u)\}. \tag{8}$$

## 7 Taste Heterogeneity: The Entropic Relaxation

Real agents are not perfectly status-ranked. Following Choo and Siow [7] and Galichon and Salanié [14], add i.i.d. Gumbel taste shocks of scale $\theta > 0$. With finitely many types, write $f_{ij}$ for the surplus of a type-$i$ man with a type-$j$ woman. Equilibrium couples $\pi_{ij}$ and singles $\pi_{i0}, \pi_{0j}$ satisfy

$$\pi_{ij} = \sqrt{\pi_{i0} \pi_{0j}}\, e^{f_{ij}/(2\theta)}, \tag{9}$$

the first-order conditions of the entropy-regularized problem

$$\max_{\pi \geq 0} \Big\{ \sum_{i,j} f_{ij} \pi_{ij} - \theta\, \mathrm{Ent}(\pi) \Big\}, \qquad \mathrm{Ent}(\pi) = 2 \sum_{i,j} \pi_{ij} \log \pi_{ij} + \sum_i \pi_{i0} \log \pi_{i0} + \sum_j \pi_{0j} \log \pi_{0j}, \tag{10}$$

subject to the population constraints. Writing $a_i = \sqrt{\pi_{i0}}$, $b_j = \sqrt{\pi_{0j}}$, $K_{ij} = e^{f_{ij}/(2\theta)}$, these become $a_i^2 + a_i (K b)_i = \mu_i$ and $b_j^2 + b_j (K^\top a)_j = \nu_j$, solved alternately in closed form by a generalized iterative proportional fitting procedure.

**Theorem 7.1** (Vanishing and Exploding Taste)**.** *Let the market be balanced and $f$ strictly supermodular. If every $f_{ij} > 0$, then as $\theta \downarrow 0$ the equilibrium converges to an optimal plan of (6) and $\mathsf{H} \to \mathsf{H}^*$. In the variant without singles this holds for any sign of $f_{ij}$, and as $\theta \to \infty$ the equilibrium converges to $\mu \otimes \nu$ and $\mathsf{H} \to \mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}})$ for independent $S_{\mathfrak{M}} \sim \mu_s$, $S_{\mathfrak{W}} \sim \nu_s$.*

*Sketch.* $\Gamma$-convergence of the regularized objectives on a compact feasible set. If every $f_{ij} > 0$, a man and a woman left single can be matched for a gain, so limit plans have no singles, and Theorem 6.1 fixes their status projection, on which $\mathsf{H}$ depends; for $\theta \to \infty$ the entropy term dominates and its unique minimizer over $\Pi(\mu, \nu)$ is the product measure. □

In Example 2.4 the endpoints are $\mathsf{H}^* = 0$ and $\mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}}) = \frac{1}{3}$. For height, they are $\mathsf{H}^* = 1$ and $\Phi(\Delta m/\sqrt{\sigma_{\mathfrak{M}}^2 + \sigma_{\mathfrak{W}}^2}) \approx 0.91$. Caring *less* about status produces *more* marrying up in the first case. Preferences and outcomes are different objects.

## 8 Main Result

**Theorem 8.1** (Hypergamy Is Solved)**.** *For any marriage market:*

*(a) universal hypergamy is feasible iff $\mu_s \succeq_{\mathrm{FSD}} \nu_s$;*

*(b) otherwise the minimal unmatched mass is $\mathsf{D} = \sup_t (G_{\mathfrak{W}} - G_{\mathfrak{M}})^+$;*

*(c) no strategy changes (a) or (b), and for continuous $F_{\mathfrak{M}}, F_{\mathfrak{W}}$ maxxing is zero-sum in rank;*

*(d) the equilibrium is the comonotone coupling, with index $\mathsf{H}^* = \mathrm{Leb}\{F_{\mathfrak{M}}^{-1} > F_{\mathfrak{W}}^{-1}\}$, under strictly supermodular transferable utility and under status-ranked preferences in the finite markets of Theorem 5.1;*

*(e) when taste shocks of scale $\theta$ are added to the transferable utility of (d), with finitely many types and no singles, the realized index tends to $\mathsf{H}^*$ as $\theta \downarrow 0$ and to $\mathbb{P}(S_{\mathfrak{M}} > S_{\mathfrak{W}})$ as $\theta \to \infty$.*

*Every quantity in (a)–(d) is a functional of $(\mu_s, \nu_s)$ alone, and (e) adds only $\theta$ and the surplus $f$. No other behavioral input is required.*

*Proof.* Theorems 2.1, 2.2, 4.3, 4.5, 5.1, 6.1, and 7.1. □

*Remark* 8.2 (Interpretation). When one side's status distribution overtakes the other's, measured hypergamy falls even if no one's preferences change. This matches the reversal of the gender gap in education documented by Esteve et al. [10], who report a global shift from educational hypergamy toward hypogamy.

## 9 Beyond Height: Multidimensional Status

Height is one coordinate of a longer list. Let each man carry a trait vector $x = (x_1, \ldots, x_k)$ (height, face, confidence, charisma, and so on), each coordinate a within-sex $z$-score, so that $\mu$ is now a distribution on $\mathbb{R}^k$. A requirement can combine traits in two ways: as a checklist, where every trait must clear its own bar, or as a trade-off, where strength in one trait offsets weakness in another.

**Theorem 9.1** (The Checklist Theorem)**.** *Suppose a fraction $\alpha$ of $\mathfrak{W}$ requires every trait to clear a bar, $x \in A = \{x_1 \geq \tau_1, \ldots, x_k \geq \tau_k\}$. Then at least $(\alpha - \mu(A))^+$ of $\mathfrak{W}$ is unmatched in every feasible matching, and for independent traits $\mu(A) = \prod_i \mu(x_i \geq \tau_i)$.*

*Proof.* The constrained women's neighborhood lies inside $A$, of mass $\mu(A)$. Apply the Hall bound from Theorem 2.2, exactly as in Theorem 3.4. □

**Example 9.2.** In the calibration of Section 3, six feet is $x_1 \geq 1$, so $\mu(x_1 \geq 1) \approx 0.16$. Adding a top-20% face and top-30% confidence leaves $\mu(A) \approx 0.16 \times 0.2 \times 0.3 \approx 0.01$ for independent traits. With a common pairwise correlation $\rho$ between traits (a one-factor Gaussian model), $\mu(A)$ rises to 0.032 at $\rho = 0.3$ and 0.054 at $\rho = 0.5$. For independent traits, bars multiply: every requirement added shrinks the eligible pool by its own pass rate. Correlation softens this without removing it: at $\rho = 0.5$ the face bar keeps 46% of the six-footers, and the confidence bar 73% of those left.

**Proposition 9.3** (Compensation)**.** *If instead $\mathfrak{W}$ ranks men by an index $s(x) = \beta \cdot x$, then $s$ is a scalar status functional, and Theorem 8.1 applies to it once $s$ is also defined on $\mathfrak{W}$ on the same scale. For independent standard Gaussian traits, the fraction of the top $\Phi(-c)$ of men by index who fall below $c$ on trait $i$ is*

$$1 - \frac{\Phi_2(-c, -c; r_i)}{\Phi(-c)}, \qquad r_i = \frac{\beta_i}{\|\beta\|}, \tag{11}$$

*where $\Phi_2(\cdot, \cdot; r)$ is the standard bivariate normal CDF with correlation $r$.*

*Proof.* $(x_i, s/\|\beta\|)$ is standard bivariate normal with correlation $r_i$, and the top $\Phi(-c)$ of men by index is the event $s/\|\beta\| \geq c$. □

*Remark* 9.4. With equal weights on height, face and confidence ($r_i = 1/\sqrt{3}$) and $c = 1$, take the top 16% of men by index, as many men as are six feet tall: 56% of them are under six feet. Under a checklist six feet is a gate; under a trade-off it is one input among several. The Six-Foot Theorem is a statement about the checklist.

Nothing requires the two sides to be scored on the same traits. Let $\mathfrak{W}$ rank men by an index $s_{\mathfrak{M}}$ and $\mathfrak{M}$ rank women by a different index $s_{\mathfrak{W}}$. Scores on different traits share no unit, so compare positions instead: write $U = F_{\mathfrak{M}}(s_{\mathfrak{M}}(x))$ and $V = F_{\mathfrak{W}}(s_{\mathfrak{W}}(y))$ for each person's quantile within their own side.

**Theorem 9.5** (Quantile Hypergamy)**.** *In a balanced market with continuous score distributions, if every member of $\mathfrak{W}$ requires $U \geq V + \delta$, where $0 \leq \delta \leq 1$, the minimal unmatched mass of $\mathfrak{W}$ is exactly $\delta$, whatever traits define $s_{\mathfrak{M}}$ and $s_{\mathfrak{W}}$.*

*Proof.* By the probability integral transform $U$ and $V$ are both $\mathrm{Unif}(0, 1)$. Shifting by $\delta$ as in Section 3, Theorem 2.2 gives $\mathsf{D}(\delta) = \sup_{t \in [0,1]} \big((1-t) - (1-t-\delta)^+\big)^+ = \delta$, attained at every $t \in [0, 1-\delta]$. □

*Remark* 9.6. At $\delta = 0$ the two quantile distributions coincide, so by Corollary 4.4 no strictly hypergamous matching exists. Once status means rank, a market in which everyone marries strictly up is not rare but impossible: by Lemma 4.1, every rank gained is a rank lost elsewhere.

## 10 Partner History

A common claim holds that fewer previous partners make for better relationships, either because the capacity to pair-bond gets used up or because a partner with no history has no one to compare you to. The data support an association, but not one confined to women. Among 2,654 married individuals, Busby, Willoughby and Carroll [6] find that more lifetime partners go with lower sexual quality, poorer communication and lower stability, with no significant difference between men and women. Using longitudinal data on both sexes, Smith and Wolfinger [16] find divorce risk highest with six or more premarital partners and next highest with one or two, while three to five do not differ

significantly from none besides the eventual spouse; the link survives controls for religion, values, family relationships, personality and mental health, shows no gender difference, and selection is not ruled out.

Wolfinger [21] reports the lowest divorce rates for women with zero or one premarital partners, an irregular pattern above that, and religiosity as part of the explanation for the low end; that analysis covers only women because the survey lacks full data on men's premarital sex. The neurobiology of pair bonding comes from studies of monogamous rodents [22], and its human version, in which bonding capacity is used up by previous partners, has not been established. We therefore treat partner history as a hypothesis, and treat it symmetrically.

**Definition 10.1** (Comparison)**.** A person whose previous partners had qualities $Q_1, \dots, Q_n$, drawn i.i.d. from a distribution $F$ with finite mean, values a new partner of quality $q$ at $q - \lambda(M_n - q)^+$, where $M_n = \max_{i \le n} Q_i$, $M_0 = -\infty$, and $\lambda \ge 0$ measures how much comparison matters. Only an ex who was better than the new partner costs anything.

**Proposition 10.2** (Diminishing Comparisons)**.** *The expected regret $R_n(q) = \mathbb{E}[(M_n - q)^+]$ is increasing and concave in $n$, with $R_0 = 0$ and*

$$R_{n+1}(q) - R_n(q) = \int_q^\infty F(t)^n \big(1 - F(t)\big)\, dt. \tag{12}$$

*For standard Gaussian quality and an average new partner ($q = 0$), $R_n = 0,\ 0.40,\ 0.68,\ 0.89,\ 1.17,\ 1.54,\ 1.87$ at $n = 0,\ 1,\ 2,\ 3,\ 5,\ 10,\ 20$.*

*Proof.* $R_n(q) = \int_q^\infty \mathbb{P}(M_n > t)\, dt = \int_q^\infty \big(1 - F(t)^n\big)\, dt$, which is finite because $F$ has a finite mean. Subtract consecutive terms; the integrand $F^n(1 - F)$ is nonnegative and decreasing in $n$ because $0 \le F \le 1$. □

*Remark* 10.3. The first partner sets the bar and each later one raises it by less, so the largest single step is from none to one, though later partners still add most of the total: the first supplies 45% of $R_3$ and 21% of $R_{20}$. This fits Wolfinger's jump from one premarital partner, usually the husband, to two [21], but not his higher divorce rate at two than at three to nine, nor Smith and Wolfinger [16], whose results suggest a link that grows stronger, not weaker, as partners accumulate. The model cannot tell comparison apart from selection, and nothing in it refers to sex: a man with many previous partners compares too. If partner history enters a status index, Theorem 4.5 applies unchanged: it re-sorts who ranks where, and every rank one person gains, someone else loses.

## 11 Abusive Partners

Definition 10.1 charges nothing for a bad ex, since only an ex who was better than the new partner costs anything. An abusive ex can still cost something by being remembered as better than they were. "Narcissistic abuse" is not a clinical diagnosis, and the studies below rate a partner's traits through the person they abused. Among 436 current and former partners and relatives of people high in narcissistic traits, Day, Townsend and Grenyer [8] find abuse alongside mutual idealization and devaluation. Among 75 women who had just left abusive partners, Dutton and Painter [9] find attachment to the former partner predicted by how extreme and intermittent the maltreatment was and by the imbalance of power, both at separation and six months later. Among 1,294 people recruited through online communities about narcissism and abuse, 94% of them women, Arabi [2] finds that a former partner's narcissism predicts symptoms of post-traumatic stress, which include intrusive memories, beyond the frequency of the abuse and any earlier abuse.

**Definition 11.1** (Idealization)**.** A previous partner is *idealized by* $\iota \geq 0$ if they are remembered at quality $Q_i + \iota$. With one idealized partner the comparison point is $M_n^\iota = \max(Q_1 + \iota, Q_2, \ldots, Q_n)$, and since the $Q_i$ are i.i.d. it does not matter which partner it was. Write $R_n^\iota(q) = \mathbb{E}[(M_n^\iota - q)^+]$.

**Proposition 11.2** (Idealization Compounds)**.** *$R_n^\iota(q)$ is nondecreasing and convex in $\iota$, with $R_n^0 = R_n$ and $R_1^\iota(q) = R_1(q - \iota)$. For standard Gaussian quality and an average new partner ($q = 0$), $R_1^\iota = \iota\,\Phi(\iota) + \varphi(\iota)$. At $\iota = 1$ this is 1.08, which is 2.7 times $R_1$ and exceeds $R_4 = 1.05$, the regret from four ordinary previous partners; at $\iota = 1.5$ it is 1.53, almost $R_{10} = 1.54$.*

*Proof.* Write $M_n^\iota = \max(Q_1 + \iota, M')$ with $M'$ free of $\iota$. It is convex and nondecreasing in $\iota$, and composing with the convex nondecreasing map $x \mapsto (x - q)^+$ preserves both properties; take expectations, which are finite because $F$ has a finite mean. Also $(Q_1 + \iota - q)^+ = (Q_1 - (q - \iota))^+$. For $Z \sim N(0, 1)$, since $\varphi'(z) = -z\varphi(z)$, $\mathbb{E}[(Z + \iota)^+] = \int_{-\iota}^{\infty} (z + \iota)\varphi(z)\,dz = \varphi(\iota) + \iota\,\Phi(\iota)$. □

Partners have diminishing returns (Proposition 10.2) and idealization has increasing ones, so abuse acts through $\iota$ and $\lambda$ rather than $n$: one ex is enough. The expected cost of comparison is $\lambda R_n^\iota(q)$, so if abuse also raises $\lambda$, as the attachment findings suggest, it multiplies all of the above. No study measures $\iota$. Nothing in the model refers to sex, although the samples in [9] and [2] are entirely or almost entirely women.

*Remark* 11.3 (First Partners). Since it does not matter which partner was idealized, what is special about a first partner lies outside the model. While that relationship lasts, $M_0 = -\infty$: there is no one to compare them with. Abuse may also lower the floor. Thibaut and Kelley's comparison level [19], the standard a relationship is judged against, is learned from experience, and Definition 10.1 has a ceiling, $M_n$, but no floor. Among 5,681 adolescents aged 12 to 18, dating violence was followed five years later by 1.9 to 3.6 times the adjusted odds of being a victim of partner violence, in both sexes, and psychological violence alone roughly doubled them [11]. This fits a lowered floor, but selection is not ruled out.

*Remark* 11.4 (Recovery). None of this need be permanent. In [9], attachment to the former partner fell by about 27% over the six months, so $\lambda$ is better read as decreasing in the time since the relationship ended. An inability to experience loving feelings is a listed symptom of post-traumatic stress disorder [1], not the used-up bonding capacity of Section 10. After breakups in general, not only abusive ones, sex to cope with distress or to get back at the ex is most frequent soon after the breakup and then declines [3].

*Remark* 11.5 (Flipping). A survivor is sometimes said to flip hypergamy and seek a partner of lower status. That is hypergamy for the status functional $-s$, so every theorem above applies with $-s$ in place of $s$, and by Theorem 2.2 the deficit becomes $\sup_t(\nu(s \leq t) - \mu(s \leq t))^+$. In the calibration of Section 3, "he must be taller than me" is feasible for everyone, but if all of $\mathfrak{W}$ requires "he must not be taller than me," at least 0.66 of $\mathfrak{W}$ is unmatched. Every union in which one partner marries down is one in which the other marries up. Whether survivors flip at all is not established. The nearest evidence concerns dominance rather than status, and depends on where the danger comes from: Colombian women who perceive more risk of domestic violence prefer less masculine Colombian faces [5], while American women more afraid of crime prefer more aggressive and formidable mates [17].

## 12 Conclusion

Hypergamy reduces to optimal transport on the real line. Its feasibility is stochastic dominance, its failure is the Hypergamy Deficit, its equilibrium is the quantile coupling, and its realized rate is the measure of the set where one quantile function exceeds another. Section 9 extends the deficit to several traits, where checklist requirements multiply when traits are independent and trade-offs reduce to a

scalar index, and Section 10 treats partner history as a comparison effect that applies to both sexes. Section 11 models an abusive ex as an idealized one, whose cost compounds where that of additional partners diminishes. Open problems include general multidimensional status (Brenier maps and upper sets in a partial order), search frictions, and the case in which the status functional is itself endogenous.

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